MATD12 – REPRESENTATION THEORY

OLLE RENÉRIUS REHNQUIST

1. Lecture 1

1.1. **Setting the scene.** In many practical situations where we invoke groups in mathematics, it is not the group that is of importance but rather how it acts on other objects. Representation theory aims to ask the question: "What are all **things** that a given group can act on?" There are a few possible candidates for **things**. We will focus on vector spaces. Fix a field k.

Given a group G and a vector space V, we say that a homomorphism $\rho : G \to \mathrm{GL}(V)$ is a *representation* of G in V. The *degree* of ρ is defined to be $\dim(V)$ which may or may not be infinite.

An important remark is that we will often fix a basis of V and consider ρ as a homomorphism from G to the matrix group $GL_n(k)$ where n is the dimension of V. We will use these definitions interchangeably and without prior warning.

Whenever one defines an object in mathematics, one is morally obligated to also introduce a notion of morphisms. Given two representations $\rho : G \to \mathrm{GL}(V)$ and $\rho' : G \to \mathrm{GL}(W)$ we say that a linear map $\varphi : V \to W$ is a *morphism of representations* if for each $g \in G$ the following diagram commutes:

$$\begin{tikzcd} V \arrow[r, "\rho_g"] \arrow[d, "\varphi"'] & V \arrow[d, "\varphi"] \\ W \arrow[r, "\rho'_g"'] & W \end{tikzcd}$$

In particular, if φ is bijective, we shall say that ρ and ρ' are *isomorphic* representations.

Example 1.1. We will now give a non-example to show that an isomorphism of representations is stronger than an isomorphism of the underlying vector space. Consider the two representations on $\mathbb{C}$ of $C_2 = \langle x \mid x^2 = 1 \rangle$ given by

$$\rho_1 : x \mapsto 1 \qquad \text{and} \qquad \rho_2 : x \mapsto -1.$$

Notice that the underlying vector spaces are the same, but suppose for contradiction that there exists an isomorphism of representation $\varphi : \mathbb{C} \to \mathbb{C}$. There exists a nonzero $\lambda \in \mathbb{C}$ such that $\varphi = \lambda I$. Since φ was assumed to be a G-homomorphism it must be the case that

$$\begin{gathered} \rho_1(x)(\varphi(1)) = \varphi(\rho_2(x)1) \iff \\ \lambda = -\lambda \end{gathered}$$

which is a contradiction. Thus, these two representations are not isomorphic.

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1.2. **Representations as modules.** For the reader that knows what a module is, we see that the definition of a morphism above is indeed a natural one. Notice that given a group G we can form the vector space $k[G]$ for which the basis consists of all elements of G. We can define a multiplication on $k[G]$ by extending the group multiplication linearly to make $k[G]$ into an algebra. Then, we see that we can identify representations with $k[G]$-modules. Given a representation $\rho : G \to \mathrm{GL}(V)$ we give V the structure of a $k[G]$-module by imposing that

$$g \cdot v = \rho_g(v).$$

Similarly, given a $k[G]$-module V, we define a homomorphism

$$\begin{aligned} \rho : G &\longrightarrow \mathrm{GL}(V) \\ g &\longmapsto (v \mapsto g \cdot v). \end{aligned}$$

Under this identification, a morphism of representations is simply a homomorphism of $k[G]$-modules.

This perspective of representations as $k[G]$-modules will be important later on.

1.3. **First examples and further definitions.**

Example 1.2. Let us look at some first few examples.

(a) The map $\rho : \mathbb{R} \to \mathbb{R}^\times$ given by $\rho_t = e^t$ is a 1-dimensional representation of $\mathbb{R}$.
(b) The map $\rho : \mathbb{R} \to \mathrm{GL}(C(\mathbb{R}))$ given by $\rho_t(f)(x) = f(x+t)$ is an infinite-dimensional representation of $\mathbb{R}$.
(c) A representation of $\mathbb{Z}$ is simply an invertible linear operator $T : V \to V$.
(d) A rperesentation of $\mathbb{Z}/n\mathbb{Z}$ is a linear operator $T : V \to V$ such that $T^n = \mathrm{id}$.

Given a representation $\rho : G \to \mathrm{GL}(V)$ and a subspace $W \subseteq V$ for which $\rho_g(W) \subseteq W$ for all $g \in G$ we say that $\rho\,|_W : G \to \mathrm{GL}(W)$ is a subrepresentation of V. From the module point of view a subrepresentation is simply a $k[G]$-submodule.

Exercise 1.3. Prove that $\rho_g(W) \subseteq W$ for all $g \in G$ is equivalent to $\rho_g(W) = W$ for all $g \in G$. If we fix $g \in G$, is it true that $\rho_g(W) \subseteq W$ if and only if $\rho_G(W) = W$? What if W is finite-dimensional?

A nonzero representation $\rho : G \to \mathrm{GL}(V)$ is called *irreducible* if it has no proper nonzero subrepresentations.

Example 1.4. Consider the following examples.

(a) We can now ask whether the representation $\rho : \mathbb{R} \to \mathrm{GL}(C(\mathbb{R}))$ given by $\rho_t(f)(x) = f(x+t)$ is irreducible. The answer is obviously no. For instance, the subspace of 1-periodic functions is a subrepresentation. Let us determine all the 1-dimensional subrepresentations of ρ. If $f(x) \in C(\mathbb{R})$ generates a 1-dimensional subrepresentation, then there exist constants $\lambda(t)$ for each $t \in G$ such that $f(x+t) = \lambda(t) \cdot f(x)$ and so in particular $f(t) = \lambda(t) \cdot f(0)$. If $f(0) = 0$ then $f \equiv 0$ and thus does not generate a 1-dimensional subspace. If, on the other hand, $f(0) \neq 0$, then $\lambda(t) = f(t)/f(0)$ is continuous and satisfies that $\lambda(t+t') = \lambda(t)\lambda(t')$. One of the later exercises implies that $f(t) = f(0) \cdot e^{at}$ for some $a \in \mathbb{R}$. It is easy to check that all functions on this form do indeed generate a 1-dimensional subrepresentation and our characterisation is therefore completed.

(b) Which representations $\rho : \mathbb{Z} \to \mathrm{GL}_n(\mathbb{C})$ are irreducible? If $n = 1$, then ρ is certainly irreducible. We claim that if $n > 1$, then ρ cannot be irreducible. Indeed, we recall from linear algebra that ρ_1 will have an eigenvalue.

(c) Consider the representation $\rho : S_3 \to \mathrm{GL}_2(\mathbb{R})$ which is given by letting S_3 act on the vertices of a regular triangle in the plane. Clearly this action does not preserve any lines through the origin and is hence irreducible.

Consider now the complexification $\rho^{\mathbb{C}} : S_3 \to \mathrm{GL}_2(\mathbb{C})$. A more interesting question is whether this representation is irreducible. Suppose for the sake of contradiction that there exists a S_3-invariant 1-dimensional subspace $\langle v \rangle \subseteq \mathbb{C}^2$. Let $r \in S_3$ correspond to a rotation and $s \in S_3$ correspond to a reflection. Since $\rho_r^3 = \mathrm{id} = \rho_s^2$ we get that both ρ_r and ρ_s are diagonalisable and since $\rho_r \neq \mathrm{id}$ we deduce that there exists a third root of unity ϵ such that $\rho_r(v) = \epsilon v$ and $\rho_s(v) = \pm v$. This gives us that $\rho_{rs}(v) = \pm\epsilon v$, but rs corresponds to a reflection, so $\pm 1 v = \pm\epsilon v$. Thus $v = 0$, a contradiction. Hence $\rho^{\mathbb{C}}$ is irreducible.

Exercise 1.5. Prove that any continuous 1-dimensional representation of $\mathbb{R}$ is given by $f(t) = e^{at}$ for some $a \in \mathbb{R}$.

Exercise 1.6. Let us expand on an example above.

(a) Prove that any linear operator $A : V \to V$ where $\dim_{\mathbb{C}}(V) > 1$ has an eigenvalue.

(b) Show that the representation $\rho : \mathbb{Z} \to \mathrm{GL}_2(\mathbb{R})$ given by
$$\rho_1 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$
is irreducible.

(c) Tweak the proof above, to show that any representation $\rho : \mathbb{Z} \to \mathrm{GL}_3(\mathbb{R})$ is reducible.

(d) We can do better. Show that for any $n > 2$, a representation $\rho : \mathbb{Z} \to \mathrm{GL}_n(\mathbb{R})$ is reducible.

1.4. **Maschke's theorem and indecomposable representations.** Our goal is to eventually be able to understand all representations of a given group (for sufficiently nice groups and over sufficiently nice fields). In a perfect world, we would be able to say that the study of irreducible representations lets us understand arbitrary representations completely. It turns out that we live in an (almost) perfect world.

Example 1.7. For a more serious example, we will consider the real representations of the dihedral group
$$D_8 = \langle r, s \mid r^4 = s^2 = 1, srs = r^{-1} \rangle.$$
We first find all the 1-dimensional representations. Suppose that $\phi : D_8 \to \mathbb{R}^\times$ is a representation. From the relations above one deduces that $\phi(r)^2 = \phi(s)^2 = 1$. Conversely each choice of $\phi(r)$ and $\phi(s)$ as ± 1 gives a representation of degree 1.

We also notice that there is degree 2 representation coming from the action of D_8 on the square situated in the plane. This representation $\rho : D_8 \to \mathrm{GL}(\mathbb{R}^2)$ can be given by
$$\rho(r) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad \rho(s) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$
Can we find any more representations of D_8? The answer is yes for a silly reason. Think of the action of D_8 on a square situated in the xy-plane of $\mathbb{R}^3$. Then this new representation

can be given by

$$\rho'(r) = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{and} \quad \rho'(s) = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Repeating this trick, we can construct arbitrarily large representations.

In the last example, we saw that we can "glue together" representations. This motivates the following definition. Given two representations $\rho_1 : G \to \mathrm{GL}(V)$ and $\rho_2 : G \to \mathrm{GL}(W)$ we can form their *direct sum* $\rho_1 \oplus \rho_2 : G \to \mathrm{GL(V \oplus W)}$ defined by

$$\rho(s)(v, w) = (\rho_1(s)v, \rho_2(s)w).$$

If ρ is a representation of G such that there do not exist proper subrepresentations ρ_1 and ρ_2 with $\rho_1 \oplus \rho_2 \cong \rho$ we say that ρ is *indecomposable*.

Exercise 1.8. Show that any irreducible representation is indecomposable.

Example 1.9. Let us consider two cases where the notion of irreducible and indecomposable representations differ.

(a) Consider the representation of $C_2 = \langle x \mid x^2 = 1 \rangle$ over the field $\mathbb{F}_2$ given by $\rho : C_2 \to \mathrm{GL}(\mathbb{F}_2^2)$ and

$$\rho(x) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

This is a representation since

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

It is **not irreducible** since the subspace generated by $(1, 0)$ gives a subrepresentation. However, one can show that this is the only subrepresentation of ρ and thus ρ is **indecomposable**.

(b) Consider the related representation $\rho : \mathbb{Z} \to \mathrm{GL}_2(\mathbb{C})$ given by

$$\rho_1 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Similarly to above, one can show that ρ_1 is not irreducible, yet indecomposable.

This is troubling, since understanding all representations is equivalent to understanding the indecomposable representations. On the other hand, irreducible representations will be easier for us to deal with. These two notions converge when G is finite and $\mathrm{char}(k) = 0$.

Theorem 1.10 (Maschke)**.** *Let k be a field of characteristic* 0 *and G a finite group. If $\rho : G \to \mathrm{GL}(V)$ is indecomposable, then it is irreducible.*

We will first give a proof of a geometric flavour that only works when $k = \mathbb{R}$ or $\mathbb{C}$ and V is finite-dimensional.

Proof. We will prove the contrapositive. Suppose that $W \subseteq V$ is a proper nonzero subrepresentation. Since V is finite-dimensional over $\mathbb{R}$ or $\mathbb{C}$ it admits an inner product B. Using this inner-product we could define $W^\perp$ and then clearly $V = W \oplus W^\perp$ but the problem is that $W^\perp$ need not be a subrepresentation (i.e. G-invariant). To fix this, we will use an averaging trick. Consider the inner product $B' = \sum_{g \in G} B(\rho_g v, \rho_g w)$ which is well-defined since G is finite. Clearly $B'(\rho_g v, \rho_g w) = B'(v, w)$ for all $g \in G$ (the terms in the sum are simply permuted). Now, let $W^\perp$ be the orthogonal complement of W with respect to this inner product B'. Then, it follows that $W^\perp$ is G-invariant since for any $v \in W^\perp$ and any $w \in W$:

$$B'(\rho_g v, w) = B'(v, \rho_{g^{-1}} w) = 0$$

since W is G-invariant. Thus, $V = W \oplus W^\perp$ is a reducible representation. The theorem follows. □

Exercise 1.11. Deduce the following for any finite group G using our geometric proof of Maschke's theorem:

(a) If $\rho : G \to \mathrm{GL}_n(\mathbb{R})$ is a representation, then there exists a basis of $\mathbb{R}^n$ such that $\mathrm{Im}\rho \subseteq O_n$.
(b) If $\rho : G \to \mathrm{GL}_n(\mathbb{C})$ is a representation, then there exists a basis of $\mathbb{C}^n$ such that $\mathrm{Im}\rho \subseteq U_n$.

Exercise 1.12. Explain why any finite-dimensional representation of a finite group over a field of characteristic 0 can be decomposed as a direct sum of irreducible representations. Does the same result hold for infinite-dimensional representations? For infinite groups? For fields of positive characteristic?

2. Lecture 2

2.1. **A fun exercise.** Recall the complex representation

$$\rho : \mathbb{R} \longrightarrow \mathrm{GL}(C(R))$$
$$\rho_t(f)(x) = f(x+t).$$

Last time, we showed that all 1-dimensional subrepresentations are given by $\langle e^{ax} \rangle$. For 2-dimensional subrepresentations, we can for instance find $\langle e^{ax}, xe^{ax} \rangle$.

Exercise 2.1. Is it true that the only n-dimensional indecomposable subrepresentation is

$$\langle e^{ax}, xe^{ax}, x^2 e^{ax}, \dots, x^{n-1} e^x \rangle.$$

2.2. **A second proof of Maschke's theorem.** We will now give a proof for Mascke's theorem (Theorem 1.10) that works in general.

Proof. Let G be a finite group, k be a field of characteristic 0 and $\rho : G \to \mathrm{GL}(V)$ a representation. Suppose that $W \subseteq V$ is a subrepresentation.

From linear algebra, we know that there exists a projection $\pi' : V \to V$ such that $\pi'(V) \subseteq W$ and $\pi'|_W = \mathrm{id}_W$. We will first try to find a projection which is a morphism of representations. This can be done by an averaging trick. Consider the linear map $\pi : V \to V$ given by

$$\pi(v) = \frac{1}{|G|} \sum_{g \in G} \rho_g \circ \pi'(\rho_{g^{-1}} v).$$

This is clearly a projection and a morphism of representations from $V \to V$. You will show in the exercises that the category of representations is abelian so $W' = \ker(\pi)$ is a subrepresentation of V and since π is a projection $W \oplus W' \cong V$ which proves that V is decomposable. This finishes the proof. □

Exercise 2.2. In this exercise, we will verify the computations in the above proof that were left out.

(a) Prove that if $W \subseteq V$ are vector spaces then there exists a projection $\pi' : V \to V$ such that $\pi'(V) \subseteq W$ and $\pi'|_W = \mathrm{id}_W$.
(b) Prove that π above is a projection.
(c) Prove that π is a morphism of representations.

Exercise 2.3. It turns out that the category of representations of a group G over a field k is abelian. You are not expected to know what this means. Instead, prove that if φ is a morphism of representations, then $\ker(\varphi)$ and $\mathrm{im}(\varphi)$ are representations.

2.3. **Schur's lemma.** In this section, our treatment will deviate somewhat from the lecture. The main theorem of this sections is the following.

Theorem 2.4 (Schur). *Let (V, ρ) and (V', ρ') be representations of a group G over a field k and let $\varphi : V \to V'$ be a morphism of representations. Then:*

(i) if V is irreducible, then φ is injective.
(ii) if V' is irreducible, then φ is subjective.
(iii) if $(V, \rho) = (V', \rho')$ is irreducible and k is algebraically closed, then $\varphi = \lambda I$ for some $\lambda \in k^\times$.

Proof. Part (i) follows from the fact that $\ker(\varphi)$ is a subrepresentation of V and part (ii) follows from the fact that $\mathrm{im}(\varphi)$ is a subrepresentation of V' using Exercise 2.3.

For part (iii), assume that $\varphi : V \to V$ is a nonzero endomorphism of an irreducible representation over an algebraically closed field k. Since k is algebraically closed, there exists an eigenvalue $\lambda \in k$ of φ. In particular, $\varphi - \lambda \cdot \mathrm{id} : V \to V$ is a morphism of V which is not injective. Hence, by part (i), $\varphi - \lambda \cdot \mathrm{id} = 0$ which finishes the proof. $\square$

What is the upshot of Schur's lemma? Let V and W be representations of a group G over a fixed base field k. We will let $\mathrm{Hom}_G(V, W)$ denote the vector space of G-homomorphisms from V to W and $\mathrm{End}_G(V)$ the $k[G]$-algebra of G-endomorphisms of V. Then:

- If V and W are nonisomorphic representations, then $\dim \mathrm{Hom}_G(V, W) = 0$.
- If $V \cong W$, then $\dim(\mathrm{Hom}_G(V, W)) > 0$.
- If V is irreducible, then the k-algebra $\mathrm{End}_G(V)$ is a division algebra. This means that every nonzero element is invertible. If k is algebraically closed, then $\mathrm{End}_G(V) \cong k$.

In the next few lectures, we will see that the very successful theory of characters is essentially a corollary of Schur's lemma. In particular, we will see that $\dim(\mathrm{Hom}_G(-, -))$ gives us a way to take an inner product of representations.

Example 2.5. As a last example, we will observe that if k is not algebraically closed, then $\mathrm{End}_G(V)$ need not be a 1-dimensional vector space over k, even if V is irreducible.

Consider the representation (V, ρ) given by

$$\rho : C_3 \longrightarrow \mathrm{GL}_2(\mathbb{R})$$
$$1 \longmapsto A = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}$$

which is a representation over $\mathbb{R}$ since $A^3 = \mathrm{id}$. This representation is irreducible since the eigenvalues of A over $\mathbb{C}$ are nonreal. What does $\mathrm{End}_{C_3}(V)$ look like? It is the collection of matrices $B = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(\mathbb{R})$ such that $BA = AB$, which are exactly the matrices on the form

$$\begin{pmatrix} a & -b \\ b & a-b \end{pmatrix} = a \cdot \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + b \cdot \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}.$$

Thus,

$$\mathrm{End}_{C_3}(V) = \left\langle \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix} \right\rangle$$

has dimension 2. As a $\mathbb{R}$-algebra,

$$\mathrm{End}_{C_r}(V) \cong \mathbb{R}[e^{2\pi i/3}].$$

What is going on here? Recall that this representation is equivalent to a $\mathbb{R}[G]$-module V. When we tensor this with $\mathbb{C}$ we get a $\mathbb{C}[G]$-module $\mathbb{C} \otimes V$ which is decomposable as the direct sum of the representations $1 \mapsto e^{2\pi i/3}$ and $1 \mapsto e^{4\pi i/3}$. The lack of these two elements in the field $\mathbb{R}$ is what is making the situation more complicated.

Exercise 2.6. Let n be a positive integer.

(a) What are the irreducible representations of $\mathbb{Z}/n\mathbb{Z}$ over $\mathbb{C}$?
(b) Let p be a prime. What are the irreducible representations of $\mathbb{Z}/p\mathbb{Z}$ over $\mathbb{Q}$?
(c) What are the irreducible representations of $\mathbb{Z}/4\mathbb{Z}$ over $\mathbb{Q}$?